# Solving Problems Using the Outcomes of Experiments

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# Event

An **event** is a particular outcome or a group of outcomes that we are interested in observing.

Example: When rolling a die, the event "rolling an even number" is the group of outcomes $2, 4, 6$.

# Frequency

The **frequency** of an outcome is the number of times it occurs during an experiment.

Example: A coin is tossed $10$ times and lands on heads $6$ times.

$$\text{Frequency of heads} = 6$$

# Relative Frequency

The **relative frequency** of an outcome is the ratio of the number of times the outcome occurs to the total number of trials.

$$\text{Relative Frequency} = \frac{\text{Frequency of the Outcome}}{\text{Total Number of Trials}}$$

# Relative Frequency: Example

A coin is tossed $10$ times and lands on heads $6$ times.

$$\text{Relative Frequency of heads} = \frac{6}{10} = \frac{3}{5}$$

# Experimental Probability

The **experimental probability** of an event is estimated using the results of an actual experiment.

$$P(E) = \frac{\text{Number of Times Event } E \text{ Occurs}}{\text{Total Number of Trials}}$$

# Quick Tip

Experimental probability is the same idea as relative frequency of an event.

Remember the three steps:

**Count** the event.
**Compare** it to the total trials.
**Convert** to a fraction, decimal, or percent.

# Ways to Write the Probability

Experimental probability can be expressed as a **fraction**, **decimal**, or **percentage**.

$$\frac{3}{5} = 0.6 = 60\%$$

# Decimal to Percentage

To convert a decimal to a percentage, multiply it by $100\%$.

$$0.6 \times 100\% = 60\%$$

$$0.25 \times 100\% = 25\%$$

# Making Predictions

Experimental results can be used to **make predictions**.

A coin lands on heads $18$ times in $30$ tosses.

$$P(\text{heads}) = \frac{18}{30} = \frac{3}{5}$$

# Making Predictions: Next Step

In $100$ similar tosses, we might predict:

$$\frac{3}{5} \times 100 = 60$$

We predict about $60$ heads.

# Prediction Formula

$$\text{Predicted Number} = P(E) \times \text{No. of Future Trials}$$

# Estimates Can Differ

Experimental probability is an **estimate**.

Actual results may differ from predictions, especially when the number of trials is small.

More trials usually give a better estimate.

# Summary

| Term | Meaning |
|---|---|
| Event | Outcome(s) we want |
| Frequency | Number of times it occurs |
| Relative frequency | Frequency $\div$ Trials |
| $P(E)$ | Estimate from an experiment |

# Examples

# Example 1

A coin is tossed $40$ times and lands on heads $24$ times. Find the experimental probability of heads.

$$P(\text{heads}) = \frac{24}{40} = \frac{3}{5} = 60\%$$

# Example 2

A die is rolled $50$ times. The number $3$ appears $10$ times. Find the experimental probability of rolling a $3$.

$$P(3) = \frac{10}{50} = \frac{1}{5} = 20\%$$

# Example 2: Prediction

Predict how many times a $3$ would appear in $200$ rolls.

$$\frac{1}{5} \times 200 = 40$$

We predict about $40$ times.

# Example 3

A spinner is spun $80$ times. It lands on red $20$ times. Find the experimental probability of red.

$$P(\text{red}) = \frac{20}{80} = \frac{1}{4} = 25\%$$

# Example 3: Prediction

Predict how many times red would appear in $400$ spins.

$$\frac{1}{4} \times 400 = 100$$

We predict about $100$ times.

# Practice Exercises

# Practice 1

A coin is tossed $50$ times and lands on heads $28$ times. Find the experimental probability of obtaining heads. Express your answer as a fraction in simplest form and as a percentage.

# Practice 2

A die is rolled $60$ times. The number $4$ appears $12$ times. Find the experimental probability of rolling a $4$.

# Practice 3

A bag contains red, blue, and green marbles. One marble is drawn and replaced each time. After $80$ trials, a red marble is drawn $32$ times. Estimate the probability of drawing a red marble in a future trial.

# Practice 4

A student spins a spinner $100$ times. It lands on yellow $25$ times. What is the experimental probability of landing on yellow? Express your answer as a percentage.

# Practice 5

A basketball player makes $36$ successful free throws out of $45$ attempts. Based on these results, how many successful free throws would you expect in $75$ attempts?

# Practice 6

A box contains small cards numbered $1$ to $5$. A card is drawn and replaced $150$ times. The number $3$ is drawn $45$ times. Find the experimental probability of drawing a $3$. Then estimate how many times a $3$ would be drawn in $300$ trials.

# Practice 7

A student records the results of rolling a die $90$ times.

| Outcome | Frequency |
|---|---|
| $1$ | $14$ |
| $2$ | $16$ |
| $3$ | $13$ |
| $4$ | $17$ |
| $5$ | $15$ |
| $6$ | $15$ |

# Practice 7 (Questions)

a. What is the experimental probability of rolling an even number?

b. What is the experimental probability of rolling a number greater than $4$?

c. Which individual outcome occurred most frequently?

# Practice 8

A school survey asks $200$ students whether they prefer basketball or volleyball. A total of $125$ students choose basketball. If the same preference rate is observed among $600$ students, how many students would you expect to prefer basketball?

# Activities

# Activity 1

A die is rolled $75$ times. The number $5$ appears $15$ times. Find the experimental probability of rolling a $5$.

# Activity 2

A student spins a spinner $120$ times. It lands on green $30$ times. What is the experimental probability of landing on green? Express your answer as a percentage.

# Activity 3

A die is rolled $150$ times, and the number $3$ appears $25$ times. Based on the experiment, predict how many times the number $3$ would appear in $240$ rolls.

# Activity 4

A coin is tossed $100$ times. It lands on tails $58$ times and heads the remaining times. Find the experimental probability of obtaining heads and tails.

# Activity 5

A school survey asks $240$ students whether they prefer badminton or volleyball. A total of $150$ students choose badminton. If the same preference rate is observed among $720$ students, how many students would you expect to prefer badminton?

# Thank you!